Cryptographic Vulnerabilities & PQC Replacements Catalog

Complete Inventory of Deprecated Algorithms • NIST FIPS 203 / 204 Migration Targets

14 LEGACY ALGORITHMS CATALOGED | **NIST FIPS STANDARDIZED** | **SHOR'S & GROVER'S MITIGATED** | **DROP-IN CODE READY**

VULNERABLE PRIMITIVE	CATEGORY	SEVERITY	CIA IMPACT	QUANTUM THREAT MECHANISM	PQC REPLACEMENT	STANDARD
RSA 1024 / 2048 / 4096	Key Exchange / Signature	CRITICAL	C I	Shor's algorithm factors moduli in O((log N)³) polynomial time on a CRQC.	**ML-KEM-768** (KEM) **ML-DSA-65** (Sig)	FIPS 203 FIPS 204
DH Diffie-Hellman	Key Exchange	CRITICAL	C	Discrete Logarithm Problem solved in polynomial time by Shor's algorithm.	**ML-KEM-768** or X25519_ML-KEM-768	FIPS 203
ECDH P-256 / P-384 / P-521	Key Exchange	CRITICAL	C	Elliptic Curve Discrete Logarithm (ECDLP) broken entirely by Shor's algorithm.	**ML-KEM-768 / 1024**	FIPS 203
ECDSA secp256r1 / secp256k1	Digital Signature	CRITICAL	I	Private key extraction via Shor's ECDLP solver allows signature forgery.	**ML-DSA-65 / 87**	FIPS 204
DSA Digital Signature Algo	Digital Signature	CRITICAL	I	Standard finite field discrete logarithm problem broken by Shor's algorithm.	**ML-DSA-65**	FIPS 204
EdDSA Ed25519 / Ed448	Digital Signature	HIGH	I	Twisted Edwards curves are vulnerable to Shor's algorithm despite modern design.	**ML-DSA-65** Alt: SLH-DSA-128	FIPS 204 FIPS 205
ECC (Generic) General Elliptic Curve	Asymmetric Ops	CRITICAL	C I	All standard elliptic curve operations fail against Shor's discrete logarithm attack.	**ML-KEM-768 / ML-DSA-65**	FIPS 203 / 204
AES-128 128-bit Symmetric Key	Symmetric Cipher	MEDIUM	C	Grover's algorithm halves brute-force security to 2^64 ops (below 112-bit floor).	**AES-256-GCM**	NIST SP 800-38D
DES 56-bit Key Single DES	Symmetric Cipher	CRITICAL	C	Broken classically; Grover's reduces key recovery to ~2^28 operations.	**AES-256-GCM**	NIST SP 800-38D
3DES / TDEA Triple-DES (112-bit)	Symmetric Cipher	HIGH	C	Sweet32 birthday attacks on 64-bit blocks; Grover's halves key to 56 bits.	**AES-256-GCM**	NIST SP 800-38D
Blowfish 64-bit Block Cipher	Symmetric Cipher	HIGH	C	Small 64-bit block size vulnerable to collision attacks; weakened by Grover's.	**AES-256-GCM**	NIST SP 800-38D
RC4 Stream Cipher (ARC4)	Stream Cipher	CRITICAL	C	Statistical keystream biases allow plaintext recovery; disallowed under RFC 7465.	**AES-256-GCM** Alt: ChaCha20-Poly1305	NIST SP 800-38D
MD5 128-bit Hash Digest	Hash Function	CRITICAL	I	Practical classical collision attacks; Grover's halves preimage security to 64 bits.	**SHA-3-256 / SHA-512**	FIPS 202
SHA-1 160-bit Hash Digest	Hash Function	HIGH	I	SHAttered collision demonstrated; Grover's halves preimage security to 80 bits.	**SHA-3-256 / SHA-512**	FIPS 202

PQC Drop-in Replacement Implementation Architecture

Automated Remediation Patterns Generated by pqc_patch_generator_tool

🔑 1. ML-KEM-768 Key Encapsulation (FIPS 203)

Replaces: RSA Key Exchange, Diffie-Hellman (DH), ECDH

```python
import oqs  # liboqs-python (Open Quantum Safe)

def pqc_key_exchange():
    kem_algo = "ML-KEM-768"  # NIST Security Level 3

    # Client: Generate public/private keypair
    with oqs.KeyEncapsulation(kem_algo) as client:
        client_pubkey = client.generate_keypair()

        # Server: Encapsulate shared secret against client public key
        with oqs.KeyEncapsulation(kem_algo) as server:
            ciphertext, secret_server = server.encap_secret(client_pubkey)

        # Client: Decapsulate ciphertext to recover identical secret
        secret_client = client.decap_secret(ciphertext)

    assert secret_client == secret_server
    return secret_client, client_pubkey, ciphertext
```

✍️ 2. ML-DSA-65 Digital Signatures (FIPS 204)

Replaces: RSA Signatures, ECDSA, DSA, Ed25519

```python
import oqs  # liboqs-python (Dilithium / ML-DSA)

def pqc_sign(message: bytes):
    sig_algo = "ML-DSA-65"  # NIST Security Level 3
    with oqs.Signature(sig_algo) as signer:
        pubkey = signer.generate_keypair()
        signature = signer.sign(message)
    return signature, pubkey

def pqc_verify(message: bytes, sig: bytes, pubkey: bytes) -> bool:
    with oqs.Signature("ML-DSA-65") as verifier:
        return verifier.verify(message, sig, pubkey)
```

🛡️ 3. AES-256-GCM Symmetric Hardening

Replaces: AES-128, DES, 3DES, Blowfish, RC4

```python
import os
from cryptography.hazmat.primitives.ciphers.aead import AESGCM

def encrypt_aes256_gcm(plaintext: bytes, aad: bytes = None):
    # 256-bit key maintains 128-bit security under Grover's algorithm
    key = AESGCM.generate_key(bit_length=256)
    nonce = os.urandom(12)  # 96-bit unique IV/nonce
    aesgcm = AESGCM(key)
    ciphertext = aesgcm.encrypt(nonce, plaintext, aad)
    return ciphertext, nonce, key

def decrypt_aes256_gcm(ciphertext: bytes, nonce: bytes, key: bytes):
    return AESGCM(key).decrypt(nonce, ciphertext, None)
```

#️⃣ 4. SHA-3-256 Cryptographic Hashing (FIPS 202)

Replaces: MD5, SHA-1 (Broken Collision Functions)

```python
import hashlib

def hash_sha3_256(data: bytes) -> str:
    # Keccak sponge construction provides 128-bit quantum
    # collision & preimage resistance against Grover attacks
    return hashlib.sha3_256(data).hexdigest()

def hash_sha512(data: bytes) -> str:
    # 512-bit digest retains 256-bit security against Grover
    return hashlib.sha512(data).hexdigest()
```

⚙️ Engineering & Operational Migration Considerations

METRIC	LEGACY CLASSICAL (E.G., RSA-2048 / ECDSA)	POST-QUANTUM REPLACEMENT (ML-KEM / ML-DSA)	ENGINEERING MITIGATION
Public Key Size	RSA: ~256 bytes • ECDSA: ~64 bytes	ML-KEM-768: **1,184 bytes** • ML-DSA-65: **1,952 bytes**	Buffer enlargement; cache public keys in memory
Ciphertext / Signature	RSA: ~256 bytes • ECDSA: ~64 bytes	ML-KEM-768: **1,088 bytes** • ML-DSA-65: **3,309 bytes**	TCP Jumbo frames; TLS 1.3 fragmented records handling
Computation Speed	RSA keygen is computationally expensive	Lattice operations (NTT) are **faster** than classic RSA	Hardware AVX2 acceleration enabled by default
Transition Strategy	Pure classical single-key	**Hybrid Scheme (X25519 + ML-KEM-768)** per NIST SP 800-227	Provides dual-layer security during transition era